RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 10/11/2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: Enterprise Information Systems		**Course Code : IM355TBB**

Note:
1. Answer all the questions

Sl. No	Questions	M	BT	CO
	Part – A			
1.	The evolution of planning systems progressed from MRP-I to _____ and then to ERP.	1	1	CO1
2.	The primary mechanism by which Enterprise Information Technology provides value to organizations is through appropriate _____ improvement.	1	1	CO1
3.	Enterprise Systems are noted for providing a tightly integrated solution to an organization's information needs. State two functions that were historically stand-alone applications.	2	2	CO2
4.	What is the Role of EIS in business decision-making? State one key function.	2	1	CO1
5.	A textile company is implementing a new EIS. Their goal is to maximize the utilization of their collective employee knowledge for project teaming and innovation. Which core EIS component is primarily responsible for managing and exploiting this intellectual capital? Justify your choice.	2	3	CO3
6.	What is the Role of EIS in business decision-making? State one key function.	2	1	CO1
	Part – B			
1	Distinguish between MRP-I and MRP-II. Explain the primary functions of each planning system and illustrate how the evolution from Material Requirements Planning (MRP-I) to Manufacturing Resource Planning (MRP-II) broadened the scope of enterprise planning.	10	2	CO2
2	Define and Differentiate the Core Components of EIS. Briefly define ERP, CRM, and SCM, and explain the unique focus of each component within the overall enterprise system framework.	10	2	CO2
3	Explain the scope of Digital Transformation in the Context of EIS. Describe what the term means for a modern enterprise and how the adoption of an integrated EIS serves as the foundational enabler for achieving digital transformation goals.	10	2	CO1
4	You are advising a high-growth tech startup. The founders are considering which EIS component to prioritize first: CRM for managing investor relations and early customer feedback, or a modular ERP for integrating the finance and HR functions. Argue for the strategic component that should be prioritized to support the startup's growth and eventual Digital Transformation.	10	3	CO3

5	What is a Business Process? Provide a comprehensive definition, and explain why identifying and defining these processes is the crucial first step before any automation or reengineering effort.	10	2	CO2

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**27th November 2024**	**Maximum Marks**	**10 + 50**
Course Code	**IM255TBB**	**Duration**	**120 Mins**
Sem	**V**	**CIE – I**	
ENTERPRISE INFORMATION SYSTEMS (Elective)			

Note:
1. Answer all the Questions.

PART – A

Sl. No	Questions	M	BT	CO
1	List two key features of a Customer Relationship Management (CRM) system.	02	L2	CO1
2	Mention two types of data commonly handled by EIS.	02	L2	CO1
3	Name two popular ERP software solutions.	02	L1	CO2
4	What is the primary function of Supply Chain Management (SCM) within EIS?	02	L1	CO1
5	What are the core modules of an ERP system?	02	L2	CO2

PART – B

Sl. No	Questions	M	BT	CO
1.	Analyse the benefits and challenges of implementing an EIS in a manufacturing organization. Support your answer with examples.	10	L2	CO1
2.	How can information technology support a company's business processes and decision making and give it competitive advantage? Give examples to illustrate your answer.	10	L3	CO3
3.	Highlight some of the issues you have faced in using SAP implemented in your institution. How you resolved those issues.	10	L3	CO2
4.	Evaluate the role of EIS in Customer Relationship Management (CRM) with specific examples of how companies have leveraged EIS for CRM.	10	L2	CO3
5.	XYZ Manufacturing Company has been facing inefficiencies in its operations due to a lack of integration between its departments. The procurement, inventory, production, and sales teams operate on standalone systems, leading to delays, data inaccuracies, and increased costs. To address these issues, the company has decided to implement an Enterprise Resource Planning (ERP) system. However, after six months of implementation, the company is still struggling with system adoption and is not seeing the expected improvements. Analyze the challenges XYZ Manufacturing is facing in implementing the ERP system. Suggest strategies to overcome these challenges and ensure successful adoption of the ERP system.	10	L4	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

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	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
Marks Distribution	Max Marks	Quiz	06	04	–	–	–	04	06	–	–	–	–
		Test	10	10	20	10	–	–	20	20	10	–	–

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 07/11/2025	**CIE – 1**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: ADVANCED DECISION MODELLING		**Course Code : IM355TBD**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is balking, reneging and jockeying?	02	L2	CO1
2.	The Probability that exactly zero units in the queuing system is _____	02	L2	CO1
3.	Write the Kendall's notation for the single server queuing system	02	L2	CO1
4.	Classify the states of Markov Chain.	02	L2	CO2
5.	In an M/M/1 queue, the arrival is Poisson with 4 per minute and the average time spent in system is 5.5 minutes for each customer. The service rate is _____ per minute	02	L3	CO1
	Part – B			
1	Customers arrive at a one-window drive according to a Poisson distribution with mean of 10 minutes and service time per customer is exponential with mean of 6 minutes. The space in front of the window can accommodate only three vehicles including the serviced one. Other vehicles have to wait outside this space. Calculate: i. Probability that an arriving customer can drive directly to the space in front of the window. ii. Probability that an arriving customer will have to wait outside the directed space. iii. How long an arriving customer is expected to wait before getting the service?	10	L3	CO1
2	At a stamp vending window of a post office 20 customer arrive on an average every 10 min. the vendor serves 5 customers in 2 min. determine: 1. Average number of customers in the system 2. Average waiting time of a customer. 3. Probability of a customer waiting for more than 3 min before being served. 4. Idle time of the vendor in a shift of 8 hours	14	L3	CO1
3	Discuss the classification of queuing models and explain how Kendall's notation helps in representing them.	06	L3	CO1

4	In a bank there is only one window. A solitary employer performs all the services required and the window remain continuously open from 7am to 1 pm. It was discovered that an average number of clients is 54 during the day and the average service time is 5min / person. Find: i. Average number of clients in the system ii. Average waiting time iii. The probability that a client has to spend more than 10 mins in the system	10	L3	CO1
5	A system has three possible states, 0, 1 and 2. Every hour it makes a transition to a different state, which is determined by a coin flip. For example, from state 0, it makes a transition to state 1 or state 2 with probabilities 0.5 and 0.5. a) Find the transition probability matrix. b) Find the three-step transition probability matrix. c) Find the steady-state distribution of the Markov chain.	10	L3	CO2

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**27th November 2024**	**Maximum Marks**	**50**
Course Code	**IM355TBD**	**Duration**	**90 Min**
Sem	**VI**	**CIE – 1**	
ADVANCED DECISION MODELLING (Professional Core Elective)			

Note:

1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part A			
1.	What is the primary objective of sequencing problems?	02	1	1
2.	How does the concept of 'setup time' impact sequencing problems?	02	3	3
3.	What is the significance of 'priority rules' in sequencing problems?	02	2	2
4.	What is the difference between 'flow shop' and 'job shop' sequencing?	02	2	2
5.	How does the concept of 'setup time' impact sequencing problems?	02	3	3
	Part B			
1	Explain the following terminologies: i. Idle time on machine ii. Total elapsed time iii. No passing rule iv. Number of machines v. Processing order	10	1	1
2	Find the sequence that minimizes the total elapsed time (in hours) required to complete the following tasks on two machines. *Task* | *A* | *B* | *C* | *D* | *E* | *F* | *G* | *H* | *I* *Machine I* | 2 | 5 | 4 | 9 | 6 | 8 | 7 | 5 | 4 *Machine II* | 6 | 8 | 7 | 4 | 3 | 9 | 3 | 8 | 11	10	2	2
3	Determine the optimal sequence of jobs that minimize the total elapsed time, based on the following information. Processing time on machines is given in hours and passing is not allowed. *Job No* | *A* | *B* | *C* | *D* | *E* | *F* | *G* *Machine* M_1 | 3 | 8 | 7 | 4 | 9 | 8 | 7 *Machine* M_2 | 4 | 3 | 2 | 5 | 1 | 4 | 3 *Machine* M_3 | 6 | 7 | 5 | 11 | 5 | 6 | 12	10	3	3

Academic year 2024-2025 (ODD Sem)

<table>
<tr><td>4</td><td>There are four jobs each of which has to be processed on machines A, B, C, D, E and F in the order ABCDEF. Processing time in hours is given in the table below. Find the optimal sequencing of jobs, minimum time required to process these jobs and the idle time for each of the machines.
<table>
<tr><td>Job</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td></tr>
<tr><td>1</td><td>15</td><td>8</td><td>6</td><td>14</td><td>6</td><td>36</td></tr>
<tr><td>2</td><td>17</td><td>7</td><td>9</td><td>10</td><td>15</td><td>22</td></tr>
<tr><td>3</td><td>21</td><td>7</td><td>12</td><td>9</td><td>11</td><td>19</td></tr>
<tr><td>4</td><td>18</td><td>6</td><td>11</td><td>12</td><td>14</td><td>17</td></tr>
</table></td><td>10</td><td>2</td><td>2</td></tr>
<tr><td>5</td><td>Two major parts P_1 and P_2 for a product require processing through machine centres. The technological sequence of the parts on six machines and manufacturing times on each machine are
<table>
<tr><td rowspan="2">Part P_1</td><td>Machine sequence:</td><td>C</td><td>A</td><td>E</td><td>F</td><td>D</td><td>B</td></tr>
<tr><td>Time (hrs):</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>1</td></tr>
<tr><td rowspan="2">Part P_2</td><td>Machine sequence:</td><td>B</td><td>A</td><td>E</td><td>F</td><td>C</td><td>D</td></tr>
<tr><td>Time (hrs):</td><td>3</td><td>2</td><td>5</td><td>3</td><td>2</td><td>3</td></tr>
</table>
What would be the optimal scheduling to minimize the total processing time for these two parts? Find also the total elapsed time. For each machine specify the part that should be processed first.</td><td>10</td><td>3</td><td>3</td></tr>
</table>

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars			CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks Distribution	Quiz	Max Marks	1	2	2	–	1	2	2	–	–	–
	Test	Max Marks	10	20	20	–	10	20	20	–	–	–

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 15th April 2026	**CIE - I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: FACILITIES PLANNING AND DESIGN		**Course Code : IM365TDA**

Note:
1. Answer all the Questions.

PART – A

SI. No.	Questions	M	BT	CO
1	State any two factors influencing plant location.	02	L2	CO1
2	Identify the suitable layout for a hospital and justify your choice.	02	L2	CO1
3	Which layout is suitable for job-order production? Give reason.	02	L1	CO2
4	Analyze any two causes of inadequate facilities planning with examples.	02	L1	CO1
5	Propose a basic facilities planning strategy for a new manufacturing firm.	02	L2	CO2

PART – B

SI. No.	Questions	M	BT	CO
1.	Explain the concept of facilities planning and discuss its significance in modern industries.	10	L1	CO1
2.	Differentiate between facilities planning process and strategic planning process in terms of meaning, scope, objective, time horizon and focus.	10	L2	CO2
3.	Explain the factors influencing plant location with suitable examples.	10	L3	CO3
4.	Recommend a suitable plant layout for a manufacturing firm and justify your choice with merits and demerits.	10	L4	CO4
5.	Potential locations A, B and C have the cost structure as shown below for producing a product expected to sell for Rs. 4250 per unit. Site | Fixed Cost/year (Rs) | Variable Cost/unit (Rs) A | 85,00,000 | 775 B | 65,00,000 | 2225 C | 70,00,000 | 1460 Find the most economical location for an expected volume of 6250 units per year.	10	L2	CO2

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DEPARTMENT OF
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Date : 16th April 2026	**CIE - 1**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Service Operations Management		**Course Code : IM365TDB**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is the primary goal of Service Operations Management?	2	1	1
2.	Name two major challenges faced by service operations managers in today's dynamic environment.	2	2	2
3.	Identify any 4 tools that have been used on you to collect information to enhance customer satisfaction.	2	3	1
4.	What type of service process is typically used in a fast-food restaurant, and why?	2	3	2
5.	How does service operations management differ from manufacturing operations management?	2	2	2
	Part – B			
1	What are the different types of service processes? Cite the example of any service organization and highlight the different processes involved in that organization.	10	2	2
2	Apply the concept of the "Poka-Yoke" (mistake-proofing) methodology to a service setting of your choice. Explain how you would design both warning and control types of poka-yokes to manage the operational risks associated with customer variability and employee variability.	10	3	2
3	Explain "Service concept" using the tabular column as applied to a "New age engineering education institute".	10	5	3
4	You are tasked with launching a new, subscription-based "tele-health" service that connects patients with doctors via video calls 24/7. This service is highly intangible and faces significant variability in demand. Create a set of success metrics that specifically measure the "moments of truth" in the virtual interaction, addressing the challenge of measuring quality for an intangible product. Hint: Go up to 3 Moments of Truth starting from Zeroth. Use tabular column and avoid writing in paragraphs.	10	4	2
5	You are a Service Operations Manager of an online retail store. What challenges would you may face as the manager and what measures would you take to overcome them?	10	3	3

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Date : 15th April 2026	**CIE - I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Design of Experiments		**Course Code : IM365TDD**

Note:
1. Answer all the Questions.
2. Use of statistical tables permitted

Sl. No	Questions	M	BT	CO
	Part – A			
1.	List two applications of design of experiments.	1	1	CO1
2.	What are the two advantages of replication?	2	1	CO1
3.	_____ is an example of larger-the-better type quality characteristic.	1	1	CO2
4.	According to Genichi Taguchi "Quality is the _____ a product causes to the society"	2	2	CO2
5.	Identify response, controllable and noise factors for gas welding process.	2	2	CO1
6.	What are the steps in the engineering design problem strategy?	2	2	CO2
	Part – B			
1	Enumerate the guidelines for the statistical approach in designing and analyzing an experiment.	10	2	CO1
2	Trace the history of statistical design. Elucidate the important stages.	10	3	CO1
3	Write the expressions for quality loss function of various quality characteristics. Illustrate with sketches. List the situations in which these quality characteristics are applicable.	10	2	CO2
4	Identify the quality control activities during various product realization steps. Comment on the ability of these activities to reduce effect of noise factors.	05	3	CO2

<table>
<tr>
<td>5</td>
<td>An industrial engineer employed by a beverage bottler is interested in the effects of two different types of 32-ounce bottles (A) on the time to deliver 12-bottle cases of the product. The two bottle types are glass and plastic. Two workers (B) are used to perform a task consisting of moving 40 cases of the product 50 feet on a standard type of hand truck and stacking the cases in a display. Four replicates of a 2^2 factorial design are performed, and the times observed are listed in the following table:
<table>
<tr><td rowspan="2">Treatment Combination</td><td colspan="4">Replicates</td></tr>
<tr><td>I</td><td>II</td><td>III</td><td>IV</td></tr>
<tr><td>1</td><td>5.12</td><td>4.89</td><td>4.98</td><td>5.00</td></tr>
<tr><td>a</td><td>6.65</td><td>6.24</td><td>5.49</td><td>5.55</td></tr>
<tr><td>b</td><td>4.95</td><td>4.43</td><td>4.27</td><td>4.25</td></tr>
<tr><td>ab</td><td>5.28</td><td>4.91</td><td>4.75</td><td>4.71</td></tr>
</table>
i. Estimate the factor effects.
ii. Prepare an analysis of variance table and determine which factors are important in explaining time to deliver.</td>
<td>15</td>
<td>2</td>
<td>CO4</td>
</tr>
</table>

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Date : 02nd May 2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Facilities Planning and Design		**Course Code : IM365TDA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Facilities planning are subdivided into the subjects of facilities _____ and facilities _____.	(02)	L1	CO1
2.	State two objectives of facilities planning.	(02)	L2	CO1
3.	List two factors of facilities location.	(02)	L1	CO1
4.	Expand CRAFT and COFAD.	(02)	L1	CO1
5.	What is the basic principle of the CORELAP algorithm, and how is the first department selected?	(02)	L1	CO1
	Part – B			
1.	Enumerate and analyse the characteristics of all the facilities of a supply chain to achieve excellence.	10	L2	CO1
2.	Apply the Winning facilities planning process methodology for facilities planning of a manufacturing organization, and provide your analysis summary.	10	L3	CO3
3.	Perform a comparative assessment of the performance characteristics of process layout, product layout and cellular layout.	10	L2	CO2

<table>
<tr>
<td>4.</td>
<td>An Auto Parts warehouse has requested that a new layout be designed for their main warehouse. The warehouse has 8 major activity centers. The current building area can accommodate 10 X 24 unit squares. Develop the new layout using ALDEP using the data given below.
<table>
<tr><th>Code</th><th>Department Name</th><th>Area (m^2)</th></tr>
<tr><td>1</td><td>Storage</td><td>12800</td></tr>
<tr><td>2</td><td>Counter</td><td>8800</td></tr>
<tr><td>3</td><td>Paint Room</td><td>9200</td></tr>
<tr><td>4</td><td>Receiving</td><td>12000</td></tr>
<tr><td>5</td><td>Office</td><td>8800</td></tr>
<tr><td>6</td><td>Parts Bins</td><td>13200</td></tr>
<tr><td>7</td><td>Tailpipe Racks</td><td>13200</td></tr>
<tr><td>8</td><td>Restroom</td><td>8000</td></tr>
</table>
<table>
<tr><td>1</td><td colspan="7">——-</td></tr>
<tr><td>2</td><td>X</td><td colspan="6">——-</td></tr>
<tr><td>3</td><td>O</td><td>O</td><td colspan="5">——-</td></tr>
<tr><td>4</td><td>A</td><td>X</td><td>U</td><td colspan="4">——-</td></tr>
<tr><td>5</td><td>U</td><td>E</td><td>U</td><td>U</td><td colspan="3">——-</td></tr>
<tr><td>6</td><td>O</td><td>A</td><td>O</td><td>U</td><td>U</td><td colspan="2">——-</td></tr>
<tr><td>7</td><td>O</td><td>I</td><td>I</td><td>U</td><td>U</td><td>I</td><td>——-</td></tr>
<tr><td>8</td><td>O</td><td>O</td><td>O</td><td>U</td><td>O</td><td>O</td><td>O</td></tr>
</table>
<table>
<tr><td>Closeness Value</td><td>A</td><td>E</td><td>I</td><td>O</td><td>U</td><td>X</td></tr>
<tr><td></td><td>64</td><td>16</td><td>4</td><td>1</td><td>0</td><td>-1024</td></tr>
</table>
Minimum Department Preference (MDP) = I = 4, Sweep Width = 2, Number of iterations to be performed, N = 3, Assume Scale: 1 square = 400 m2, Length X Width for the layout: 10 X 24</td>
<td>10</td>
<td>L3</td>
<td>CO3</td>
</tr>
</table>

5.	Given the initial layout, flow matrix and cost matrix shown in figures 1,2 and 3, use the CRAFT scoring method and procedure two ways to obtain an improved layout.(consider any two possible exchanges).	10	L1	CO1

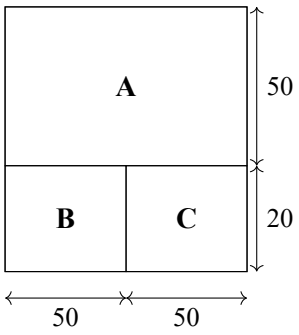


Fig 1. Initial Layout

From \ To	A	B	C
A		4	1
B	2		2
C	2	1	

Fig. 2. Flow Matrix

From \ To	A	B	C
A		1	1
B	1		1
C	1	1	

Fig. 3 Cost Matrix

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Date : 2nd May 2025	**CIE – 1**	**Max. Marks : 10 + 50**
Semester : VI	**UG**	**Duration : 30 + 90 Min**
Course Title: Service Operations Management		**Course Code : IM365TDB**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is the primary goal of Service Operations Management?	2	1	1
2.	Name two major challenges faced by service operations managers in today's dynamic environment.	2	2	2
3.	Differentiate between B2B and B2C services with one example each.	2	3	1
4.	What type of service process is typically used in a fast-food restaurant, and why?	2	3	2
5.	How does service operations management differ from manufacturing operations management?	2	2	2
	Part – B			
1	Identify any 5 tools that have been used on you to collect information to enhance customer satisfaction. Now work backwards and assess what they were trying to understand and how they might have probably improved the service offering. (Use tabular columns and showcase structured assessment)	10	2	2
2	You are a Service Operations Manager of an online retail store. What challenges would you may face as the manager and what measures would you take to overcome them?	10	3	3
3	What are the different types of service processes? Cite the example of any service organization and highlight the different processes involved in that organization.	10	2	2
4	Explain the factors that you will assess as "Service Outcomes" for a service of your choice.	10	2	2
5	Describe a service from both an operational and customer perspective. Assess the mismatches between these perspectives.	5	4	3